

Speaking about applications, Azad adds, "There is lot of effort being put into consumer robotics as the market is not completely explored. With applications like cleaning floors, watching the house when nobody is around and raising an alarm to inform the owner and authorities, telepresence robots, etc, most of our effort goes into making robots and applications more human friendly. There is a buzz among the consumers in India about how robots can be used in homes but it's going to be a long wait for our cost-sensitive Indian consumer market before we really see some major change happen, like the mobile phone revolution."

Robots in education

Perhaps one of the most exciting applications of robotics today is in education. Robotics, as a subject or hobby, helps induce a scientific temper in children from a very tender age, gradually improving their exploratory skills and multi-disciplinary knowledge. Robots also help demonstrate several concepts, especially in subjects like physics and mathematics. This helps transform education from blackboard teaching to practical learning!

Dr Venkatesh analyses: "The simplest and most effective means is to implement robotics as a hobby in school, with robotics clubs and competitions for the higher classes. The hobby effort will develop interest towards learning and the competitions will help innovation and improvise things. The stress should be on practical and hands-on activities rather than textbooks and exams. This is a thing for proper planning by the right people. Otherwise, it will turn out to be just another purely boring theory subject, with students adding robotics to their tuition list."

Some big schools are already offering robotics as a co-curricular activity on a regular basis, and this is welcomed by parents too. If the education board takes some steps to promote

robotics, it could be taken up by more schools.

At the higher education levels too, robotics education is extremely beneficial for students.

"A curriculum incorporating robotics through courses and competitions can provide practical skills and a solid grounding in mechatronics and control design theory. Robotics development requires interdisciplinary approaches that combine control design, kinematics/dynamics analysis, vision systems and trajectory planning. Integrating robotics into the curriculum provides students

the school level to postgraduate research. There are quite a few start-ups offering kits in India, not to forget the offerings from international players like LEGO and VEX Robotics.

"There are wheel bots, hexapods, bipeds and redundant robotic manipulators along with software toolkit. Price depends on the complexity and adaptability of both the robot mechanism as well as the control intelligence. The price ranges from a few thousands to a few lakhs of rupees," explains Jayakrishnan.

Shekar, however, believes that there is always a lot of satisfaction in

Present efforts towards an intelligent robotic system aim to localise the thought process in such a way that it avoids extensive processing time. One simple solution is the sense-action methodology rather than sense-process-action

with skills that they can apply in an engineering career," says Prashant Rao, technical manager-application engineering and training, MathWorks India.

A student himself, Pitchai appreciates the value of robotics: "The field can really take off by introducing small robotic workshops for students to get hands-on experience with robotic kits. At the college level, there is a need to establish robotics labs and conduct events throughout the academic year, emphasising and encouraging students to utilise the facilities and participate in competitions held at various places within the country and around the world."

The trend is already picking up, with several big institutes setting up robotics labs and conducting interesting annual fests. Well-populated online forums like Robotics India also help keep the interest in robotics alive, apart from sharing knowledge and discovering solutions to problems like unavailability or high cost of components.

There are robotic kits available for various purposes of study, right from

buying the components independently for a particular project. This is not only more economical but also offers a richer experience than simply buying a kit and following the accompanying instructions.

In general, there is immense enthusiasm amongst students, but there is a need for regular availability of guidance, components and other resources. Educational institutions need to understand both the aspects of robotics in the educational system—that is, using robotics for teaching other subjects and robotics itself as a full-fledged subject.

Robotics as an industry too offers several obvious and hidden opportunities including the making of kits, development of control software, customisation of automation systems, deployment of robot-aided systems for various industries, and so on. For this ecosystem in India to grow further, there needs to be a deeper interaction between the industry and the students. ●

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